EXP_2026Q2_INVALID_SPLIT · VERDICT REPORT · ENGINEER

GROWTH · PRICING · 2026-06-05 · 10,000 UNITS · 14 DAYS

# Invalid assignment.

*The assignment ratio drifted past the $\chi^2$ threshold. No metric value was read.*

TL;DR

**INVALID ASSIGNMENT.** **Pause and audit the assignment service before reading any metrics.**

*The observed assignment split fell outside the $\chi^2$ threshold. The lift number you'd compute would be confounded with the assignment imbalance, so the tree refuses to report it.*

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ASSIGNMENT IMBALANCE

## 4,229 / 5,771

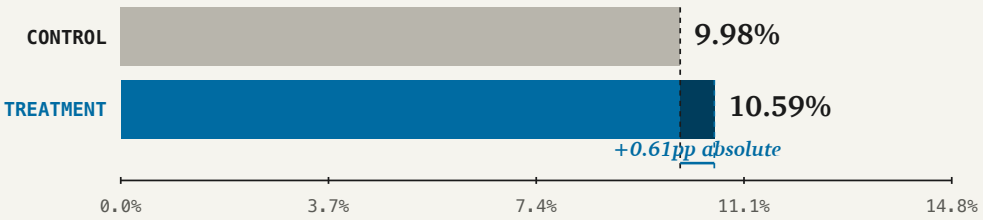
*control / treatment — expected near 50 / 50*

| $X^2$ STATISTIC | P-VALUE (SRM) |
|---|---|
| 237.78 | 0.0000 |
| THRESHOLD | DECISION STEP |
| 0.0005 | step 1 |

PRIMARY METRIC · CONVERSION_RATE

## Treatment 10.59% vs Control 9.98%

*Treatment and control rates on the same axis. The dark blue segment on the treatment bar is the absolute lift; the dotted line shows the lift relative to the control end.*

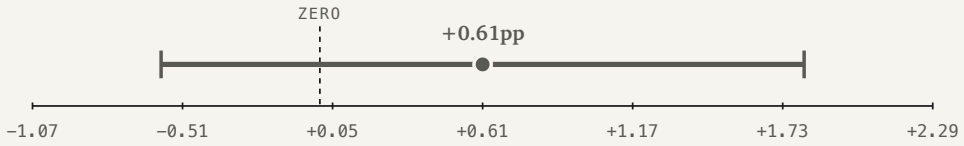



*Per-arm conversion. The lift is +0.61pp absolute (+6.1% relative).*

CONFIDENCE INTERVAL ON LIFT

## 95% CI on lift: (-0.59, +1.81)pp — straddles zero.

*Zero is inside the interval. We can't reject the null at a = 0.05. The verdict tree then asks whether we had enough sample to detect what we cared about.*




*The CI bar shows the range of plausible true lifts. The crossbars are the 95% bounds; the dot is the point estimate.*

GUARDRAILS

## Guardrail clear.

*The pre-registered guardrails are revenue and (where wired) latency. None crossed their harm thresholds.*

**revenue_per_user**

`threshold ≥ −5.0% · higher_is_better`



`90% CI [−1.17, +8.83]% · realized +3.83% ·`

CLEAR

DECISION TREE · HOW THE VERDICT WAS REACHED

## Walked 8 steps; step 1 fired the verdict.

*Each row is a checkpoint in the verdict tree. Walking stops at the row marked FIRED; that step's logic is what produced the verdict above.*

| | | |
|---|---|---|
| 0 | Required inputs present (UNVERIFIABLE guard) | `all non-null` |

| | | |
|---|---|---|
| 1 | SRM passes (or override resolved) · **FIRED** | `srm_pass = False` |
| 2 | Guardrails clear (no 90% CI on harm side) | `guard_lift = +3.83%` |
| 3 | Powered (n_observed ≥ n_required) | `n = 10,000 / 8,000` |
| 4 | Primary CI excludes 0 | `CI [−0.59, +1.81] pp` |
| 5 | Primary CI not on harm side | `direction = higher_is_better` |
| 6 | Lift magnitude ≥ MDE/2 | `\|lift\| = 0.609pp vs MDE/2 = 0.100pp` |
| 7 | Novelty stable (late_ratio ≥ 0.7) | `late_ratio = —` |

PROVENANCE · AUDITABLE CHAIN

## Reproducible from sealed brief + warehouse fixture.

*Every quantitative claim above resolves to a row in the warehouse, a function in `agentxp.stats.*`, or a step in `agentxp.interpret.tree`. The chain hash below covers the brief, every stat result, and the verdict tree's terminal step.*

```
EXPERIMENT_ID           exp_2026q2_invalid_split
CHAIN_HASH
34a07f2277a6237d52ae2a16164f8d176eb1ace2283d5d593e705192c71e411e
RENDER_STATUS           VERIFIED
AGENTXP_VERSION         0.3.0-v3
GENERATED_AT            2026-06-05T15:42:04Z
PRIMARY_METRIC          conversion_rate · proportion_test
SRM_TEST                srm_chi_squared · threshold 0.0005
TREE_MODULE             agentxp.interpret.tree.walk_tree
REPRODUCE               python scratch/walk_experiment.py
exp_2026q2_invalid_split
```